Matrices: Invariant Points and Lines

Questions 1–37

Question 1. What are the coordinates of the point that is invariant under all 2D linear transformations?

Question 2. Determine whether the point with coordinates $(2, 5)$ is invariant under each of the linear transformations represented by these matrices:

a) $\begin{pmatrix} 1 & 0 \\ 5 & -1 \end{pmatrix}$

b) $\begin{pmatrix} 6 & -2 \\ 10 & -3 \end{pmatrix}$

c) $\begin{pmatrix} 15 & -5 \\ 11 & -4 \end{pmatrix}$

d) $\begin{pmatrix} 21 & -8 \\ 100 & -39 \end{pmatrix}$

e) $\begin{pmatrix} 50 & -21 \\ 31 & -12 \end{pmatrix}$

Question 3. Determine whether the point with coordinates $(1, -2, -1)$ is invariant under each of the linear transformations represented by these matrices:

a) $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & -1 \end{pmatrix}$

b) $\begin{pmatrix} 2 & -1 & 3 \\ 3 & 4 & -3 \\ 2 & -1 & 5 \end{pmatrix}$

c) $\begin{pmatrix} 19 & 1 & 16 \\ 12 & 11 & -8 \\ -5 & 3 & -10 \end{pmatrix}$

d) $\begin{pmatrix} \frac{1}{2} & -\frac{5}{12} & \frac{1}{3} \\ -\frac{1}{3} & \frac{1}{4} & \frac{5}{6} \\ \frac{3}{2} & \frac{6}{5} & \frac{1}{10} \end{pmatrix}$

Question 4. Show that the origin is the only point that is invariant under the transformations represented by the following matrices:

a) $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$

b) $\begin{pmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{pmatrix}$

c) $\begin{pmatrix} -1 & 5 \\ -3 & 2 \end{pmatrix}$

d) $\begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{pmatrix}$

e) $\begin{pmatrix} -1 & 1 & 2 \\ -1 & 2 & 0 \\ 0 & 3 & -2 \end{pmatrix}$

f) $\begin{pmatrix} 2 & 3 & -1 \\ -3 & 1 & 2 \\ -2 & 2 & 3 \end{pmatrix}$

Question 5. Show clearly that the point $(\sqrt{5}, -4)$ is invariant under the linear transformation represented by the matrix $\begin{pmatrix} 9 & 2\sqrt{5} \\ 16 & 1+4\sqrt{5} \end{pmatrix}$.

Question 6. Find the coordinates of a point, other than the origin, that is invariant under the transformation represented by the matrix $\begin{pmatrix} \frac{3}{11} & \frac{10}{11} \\ \frac{3}{5} & \frac{1}{4} \end{pmatrix}$.

Question 7. Find the coordinates of a point, other than the origin, that is invariant under the transformation represented by the matrix $\begin{pmatrix} -\frac{1}{8} & -\frac{27}{16} \\ -\frac{1}{4} & \frac{5}{8} \end{pmatrix}$.

Question 8. Find two points that are invariant under the transformation represented by the matrix

$$\begin{pmatrix} \frac{83}{19} & -\frac{24}{19} \\ -\frac{32}{19} & \frac{31}{19} \end{pmatrix}.$$

Question 9. Find the coordinates of a point, other than the origin, that is invariant under the transformation represented by the matrix $\begin{pmatrix} 2 & 0 & 2 \\ 1 & 3 & 0 \\ 0 & 2 & -1 \end{pmatrix}$.

Question 10. Find the coordinates of a point, other than the origin, that is invariant under the transformation represented by the matrix $\begin{pmatrix} 1 & -4 & -6 \\ 4 & -3 & 2 \\ -3 & 2 & -2 \end{pmatrix}$.

Question 11. Given that $m \neq -3$, show that the point with coordinates $(1, 6)$ is an invariant point under the transformation represented by the matrix $\begin{pmatrix} \frac{10}{m+3} & \frac{m-7}{6m+18} \\ \frac{m+1}{m+3} & \frac{5m+17}{6m+18} \end{pmatrix}$.

Question 12. Find the invariant points for each of the following matrices. If there is a line of invariant points, write down its equation.

a) $\begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$

b) $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$

c) $\begin{pmatrix} 4 & 0 \\ 0 & 4 \end{pmatrix}$

d) $\begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$

e) $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$

f) $\begin{pmatrix} 1 & 6 \\ 0 & 1 \end{pmatrix}$

g) $\begin{pmatrix} 1 & 0 \\ -4 & 1 \end{pmatrix}$

h) $\begin{pmatrix} 4 & 0 \\ 0 & 0 \end{pmatrix}$

i) $\begin{pmatrix} -2 & -2 \\ 3 & 3 \end{pmatrix}$

j) $\begin{pmatrix} -3 & -3 \\ 2 & 2 \end{pmatrix}$

Question 13. It is given that $\mathbf{A} = \begin{pmatrix} 8 & 4 \\ -3 & -2 \end{pmatrix}$. Show that there are no lines of invariant points for the transformation represented by the matrix $\mathbf{A}$.

Question 14. It is given that $\mathbf{A} = \begin{pmatrix} \frac{3}{5} & \frac{4}{5} \\ \frac{4}{5} & -\frac{3}{5} \end{pmatrix}$ represents a reflection in a line that passes through the origin. Explain why the transformation must have a line of invariant points and determine the equation of this line.

Question 15. It is given that

$$\mathbf{A} = \begin{pmatrix} \frac{1-k^2}{1+k^2} & \frac{2k}{1+k^2} \\ \frac{2k}{1+k^2} & \frac{k^2-1}{1+k^2} \end{pmatrix}.$$

Show that the transformation associated with the matrix $\mathbf{A}$ has the equation $y = kx$ as a line of invariant points.

Question 16. Find the invariant points for each of the following matrices. If there is a line of invariant points, write down its equation.

a) $\begin{pmatrix} 1 & -1 & 2 \\ -1 & 0 & 3 \\ 3 & -1 & 0 \end{pmatrix}$

b) $\begin{pmatrix} -1 & 0 & 4 \\ 2 & 3 & 2 \\ 1 & 1 & 2 \end{pmatrix}$

Question 17. For each of the following 2D linear transformations, state the invariant point or find the equations of any invariant lines, explaining whether they are actually lines of invariant points:

a) Reflection in the y-axis
b) Rotation by 90° anticlockwise about the origin
c) Stretch, parallel to the x-axis, scale factor 3
d) Reflection in the x-axis
e) Rotation by 180° about the origin
f) Stretch, parallel to the y-axis, scale factor 2
g) Reflection in the line $y = x$
h) Reflection in the line $y = -x$

Question 18. For each of the following 3D linear transformations, state the invariant point, the invariant line or the invariant plane:

a) Rotation by 90°, anticlockwise about the x-axis
b) Reflection in the plane $y = 0$

Question 19. It is given that $\mathbf{A} = \begin{pmatrix} 3 & 4 \\ 2 & 5 \end{pmatrix}$.

a) Show that

$$\mathbf{A}\begin{pmatrix} 3 \\ 3 \end{pmatrix} = k\begin{pmatrix} 3 \\ 3 \end{pmatrix},$$

where k is a constant to be found.

b) Hence write down an invariant line under the transformation represented by $\mathbf{A}$.

Question 20. It is given that $\mathbf{A} = \begin{pmatrix} -2 & 9 \\ 3 & -8 \end{pmatrix}$.

a) Show that $\mathbf{A}\begin{pmatrix} -4 \\ 4 \end{pmatrix} = k\begin{pmatrix} -4 \\ 4 \end{pmatrix}$, where k is a constant to be found.

b) Hence write down an invariant line under the transformation represented by $\mathbf{A}$.

c) Show that $\mathbf{A}\begin{pmatrix} 6 \\ 2 \end{pmatrix} = p\begin{pmatrix} 6 \\ 2 \end{pmatrix}$, where p is a constant to be found.

d) Hence write down a second invariant line under the transformation represented by $\mathbf{A}$. Explain why this is a line of invariant points.

Question 21. Show that $y = 5x$ is an invariant line of the transformation given by the matrix $\begin{pmatrix} 2 & 1 \\ -5 & 8 \end{pmatrix}$.

Question 22. Show that $y = -\frac{1}{2}x$ is an invariant line of the transformation given by the matrix $\begin{pmatrix} -3 & 2 \\ 5 & 6 \end{pmatrix}$.

Question 23. Show that $y = \frac{1}{2-\sqrt{5}}x$ is an invariant line of the transformation given by the matrix $\begin{pmatrix} -7 & -4 \\ -4 & 9 \end{pmatrix}$.

Question 24. Show that $y = \frac{1}{e^2}x$ is an invariant line of the transformation given by the matrix $\begin{pmatrix} e^4 & e^{-2} \\ e^2 & e^{-4} \end{pmatrix}$.

Question 25. Show that $y = \frac{\sqrt{6}}{3}x$ is an invariant line of the transformation given by the matrix $\begin{pmatrix} \ln 2 & \ln 8 \\ \ln 4 & \ln 2 \end{pmatrix}$.

Question 26. Show that $y = \frac{1}{1+\sqrt{2}}x$ is an invariant line of the transformation given by the matrix $\begin{pmatrix} \sin k \cos k & \sin k \cos k \\ \sin k \cos k & -\sin k \cos k \end{pmatrix}$.

Question 27. Find the invariant lines for each of the following matrices. For any invariant lines found, determine whether they are lines of invariant points.

a) $\begin{pmatrix} 1 & 1 \\ -2 & 4 \end{pmatrix}$

b) $\begin{pmatrix} 6 & 2 \\ -4 & 15 \end{pmatrix}$

c) $\begin{pmatrix} 5 & -3 \\ -1 & 7 \end{pmatrix}$

d) $\begin{pmatrix} -25 & 24 \\ 12 & 3 \end{pmatrix}$

e) $\begin{pmatrix} \frac{3}{7} & -\frac{24}{7} \\ -\frac{4}{7} & -\frac{17}{7} \end{pmatrix}$

f) $\begin{pmatrix} \frac{39}{19} & -\frac{5}{19} \\ \frac{14}{19} & \frac{18}{19} \end{pmatrix}$

Question 28. Show that the transformation represented by the matrix $\begin{pmatrix} 2 & 0 \\ 1 & 1 \end{pmatrix}$ has an invariant line and a line of invariant points.

Question 29. Show that the transformation represented by the matrix $\begin{pmatrix} 3 & 1 \\ 0 & 1 \end{pmatrix}$ has an invariant line and a line of invariant points.

Question 30. It is given that $\mathbf{A} = \begin{pmatrix} 2 & -\frac{1}{2} \\ 8 & -3 \end{pmatrix}$ represents a transformation where the line $y = px$ is a line of invariant points and $y = qx + c$ is a family of invariant lines. Find the value of p and the value of q.

Question 31. Show that the transformation represented by the matrix $\begin{pmatrix} -6 & -3 \\ 5 & -1 \end{pmatrix}$ has no invariant lines.

Question 32. Show that the transformation represented by the matrix $\begin{pmatrix} -10 & 6 \\ -8 & -15 \end{pmatrix}$ has no invariant lines.

Question 33. Given $\mathbf{A} = \begin{pmatrix} \frac{24}{25} & \frac{7}{25} \\ \frac{7}{25} & -\frac{24}{25} \end{pmatrix}$ represents a reflection in a line that passes through the origin,

a) determine the equation of the line of invariant points.

b) It is given that $y = kx + 1$ is an invariant line under the transformation. State the value of k.

Question 34. The matrix $\mathbf{A} = \begin{pmatrix} 28 & -9 \\ 4 & 16 \end{pmatrix}$ represents the transformation A.

a) Given that the line with equation $y = kx$ is invariant under A, find the value of k.

b) Determine whether $y = kx$ is also a line of invariant points.

Question 35. The matrix $\mathbf{A} = \begin{pmatrix} 241 & -60 \\ 980 & -244 \end{pmatrix}$ represents the transformation A. Two lines of the form $y = kx$ are invariant under A. Show that one of these is also a line of invariant points.

Question 36. Given $\mathbf{A} = \begin{pmatrix} 2 & k \\ -3 & 4 \end{pmatrix}$, find the values of k such that the transformation represented by A has no invariant lines that pass through the origin.

Question 37. It is given that $\mathbf{A} = \begin{pmatrix} 2k & 3 \\ -1 & 2k-3 \end{pmatrix}$. Show that, irrespective of the value of k, the transformation represented by A has no invariant lines that pass through the origin.